

• CARACAS, VE • SYSTEMS ENGINEERING

Alejandro.Villa

Systems Engineering Student — USM · Full-Stack
Developer

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9th-semester Systems Engineering student at USM, Caracas. I have designed and deployed 2 systems in active production with real users: a multi-store ERP platform, an AI-powered academic repository, and a phishing threat analyzer. Master of the full development lifecycle— from relational database modeling to responsive UI design—using React, Node.js, Python, and MySQL. Looking for opportunities as a junior developer or product engineer intern.

// PROJECTS

Featured Projects

Vesta Retail ERP

[➤ Demo](#)[GitHub](#)

• Active production ERP/POS system for multi-store inventory management

Multi-store enterprise platform with distributed SPA architecture. Real-time inventory synchronization and concurrent invoicing using ultra-low latency WebSockets (Socket.io) and a relational MySQL database with Sequelize ORM optimized for robust ACID transactions.

[React 19](#)[Node.js](#)[Socket.io](#)[MySQL](#)[Sequelize](#)[Express](#)

Academic Repository + AI

[➤ Demo](#)[GitHub](#)

• School system in production – automates academic research preservation

Premium digital system for Ntra. Sra. de Fátima School. Automates preservation and intelligent analysis of scientific projects with RBAC auth (Director/Teacher), immutable auditing module, automatic Google Drive synchronization, and PDF analysis via Gemini 2.0 Flash.

[React 19](#)[Node.js](#)[Gemini 2.0](#)[MySQL](#)[JWT / RBAC](#)[Drive API](#)

PhishShield AI

[➤ Demo](#)[GitHub](#)

Metrics: 96.7% accuracy in heuristic threat detection and false positive rate reduced to 1.2%

Defensive cybersecurity platform (Software Engineering course project, USM). Integrates a custom Explainable AI heuristic classification engine, concurrent reputation checks (VirusTotal/PhishTank APIs), a high-performance concurrent multi-threaded scraper using ThreadPoolExecutor, and SQLite optimized in 4NF.

[Python 3.10+](#)[Flask](#)[SQLite 4NF](#)[VirusTotal API](#)[BeautifulSoup4](#)[Cybersecurity](#)

// STACK

Technical Skills

FRONTEND

React 19 · Vite · JavaScript ES6+ · HTML5 · CSS3 /
CSS Modules · Tailwind CSS · Framer Motion ·
Three.js

BACKEND

Node.js · Express · Python · Flask · Django · REST
APIs · WebSockets (Socket.io) · JWT

DATABASES

MySQL · PostgreSQL · SQLite (up to 4NF) ·
Sequelize · Advanced relational modeling

AI / OTHERS

Google Gemini 2.0 SDK · OpenCV · Pandas ·
BeautifulSoup4 · VirusTotal API · Google Drive API ·
Git · GitHub

// EXPERIENCE

Professional Experience

Apr 2026 – present

Software Developer

Nuestra Señora de Fátima School · Community Service

Design and implementation of a local network (LAN) Full-Stack web system for research document management. Automates academic workflows with secure JWT authentication, role-based access control, and daily backups automatically synced to Google Drive.

Dec 2025 – Apr 2026

Freelance Full-Stack Developer

Vesta Retail ERP · Independent Project

Designed, developed, and deployed to production a multi-store ERP platform that reduced estimated physical stock reconciliation time by ~35%. Automated invoicing workflows and generated daily cash closing reports synchronized in real-time between branches.

Oct 2025 – Nov 2025

Cybersecurity Software Developer (Academic)

PhishShield AI · USM Software Engineering Project

Developed an intelligent URL phishing threat analyzer. Designed and implemented a heuristic classification engine powered by Explainable AI, concurrent global reputation API queries, and a SQLite database normalized in 4NF, achieving 96.7% accuracy in tests.

// COURSES

Courses & Certifications

UNEWB, 2025

Programming Diploma — Python (Flask/Django) · JavaScript · AI for Programmers · MySQL · PostgreSQL

UCV, 2023

C++ Programming — Levels 1 & 2

CEBA UCV, 2025

Basic English

Benllisoft, 2021

Photoshop CC Level 1 — Graphic Design

// EDUCATION

Education

2022 – present

Systems Engineering

Santa Maria University (USM) · Caracas · 9th Semester

Studies in Full-Stack web development, relational database systems, software engineering, cybersecurity, and computer vision.

alejandro.dev // may 2026 — updated from ale.dev